

Architecture & Deployment

Doctor Clinic DevOps System

Architecture and Diagram Pack

Muhammed Munir Al Tawil · DevOps Course · May 2026

1. High-Level Architecture

The platform is organized into 7 layers: User, Web, Application, Database, CI/CD, Deployment, and Operations.

Layer	Component	Description
User Layer	Browser	Patients, doctors, admins via web browser
Web Layer	Nginx Alpine	Reverse proxy, port forwarding 8080 → 80
Application Layer	Laravel / PHP-FPM 8.2	Auth, dashboards, appointments, content
Database Layer	MySQL 5.7 (Docker)	Users, doctors, appointments, reports
CI/CD Layer	GitLab CI/CD	5-stage pipeline: validate → build → test → package → deploy
Deployment Layer	Proxmox VE + Docker Compose	Live production VM, public URL, NAT routing
Operations Layer	Backup + Monitoring + Terraform	Backup script, monitoring plan, IaC starter

2. Proxmox Network Architecture

The live deployment uses a two-tier network on Proxmox VE:

```
Internet
|
▼ 51.158.200.127 (Proxmox host public IP)
vmbr0 — [iptables DNAT :8080 → 10.10.10.100:8080]
|
vmbr1 (10.10.10.1/24 internal bridge + NAT masquerade)
|
▼ 10.10.10.100 (VM 100 - Doctors-Clinic-DevOps-System)
doktors_nginx :80 → doktors_app :9000 → doktors_mysql :3306
```

Component	IP / Port	Role
Proxmox Host	51.158.200.127	Public-facing server, iptables router
vmbr0	51.158.200.127/24	External bridge to internet
vmbr1	10.10.10.1/24	Internal bridge with NAT masquerade
VM 100	10.10.10.100	Ubuntu 22.04, Docker host
Nginx container	0.0.0.0:8080 → 80	Reverse proxy, static files
App container	9000/tcp (internal)	PHP-FPM Laravel runtime
MySQL container	3307:3306 (internal)	Persistent data with Docker volume

3. Docker Compose Stack

Service	Image	Port	Volume / Network
app	doctor-clinic-app (custom)	9000/tcp	./:/var/www doktors_network
nginx	nginx:alpine	8080:80	./:/var/www default.conf doktors_network
mysql	mysql:5.7	3307:3306	doktors_mysql_data (persistent volume)

4. GitLab CI/CD Pipeline

The pipeline runs 5 sequential stages on every push to the repository:

Stage	Job	Description
validate	php_syntax	PHP lint check on all source files using <code>php -l</code>
build	build_frontend	npm ci + npm run build (Vite assets), artifacts saved
test	test_application	SQLite DB setup, migrations, php artisan test (PHPUnit)
package	package_project	tar.gz archive of full project including all DevOps files
deploy	deploy_placeholder	Deployment stage (live target: Proxmox VM)

5. Kubernetes Manifests

10 Kubernetes YAML files are prepared for future cluster deployment:

- namespace.yaml – dedicated clinic namespace
- secrets.yaml – base64-encoded DB credentials
- pvc.yaml – persistent volume claim for MySQL
- mysql-deployment.yaml + mysql-service.yaml
- app-deployment.yaml + app-service.yaml
- nginx-configmap.yaml + nginx-deployment.yaml + nginx-service.yaml

6. Infrastructure as Code (Terraform)

Terraform starter files are included for future cloud infrastructure provisioning:

- provider.tf – cloud provider configuration
- main.tf – core resource definitions
- variables.tf – parameterized configuration
- outputs.tf – output values

- `terraform.tfvars.example` – example variable values